



Amazon S3

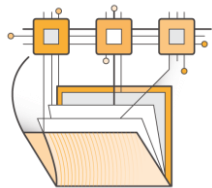
Deep Dive

Carl Summers, Software Development Engineer
Omair Gillani, Sr. Product Manager

4/19/2016

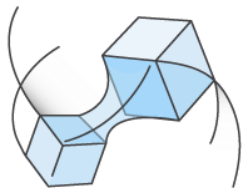


AWS storage maturity

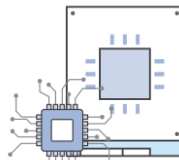


Amazon EFS

File



Amazon EBS

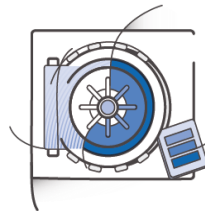


Amazon EC2
instance store

Block



Amazon S3



Amazon Glacier

Object

Data transfer



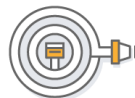
AWS Direct
Connect



Snowball



ISV connectors



Amazon
Kinesis
Firehose



Transfer
Acceleration



AWS Storage
Gateway

Our customer promise



Durable

11 9s



Available

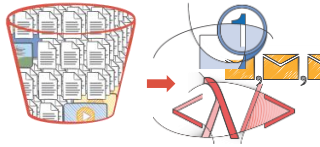
Designed for 99.99%



Scalable

Gigabytes -> Exabytes

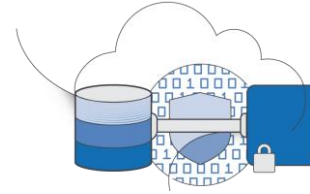
Innovation for Amazon S3



Event notifications



Cross-region
replication



VPC endpoint
for Amazon S3



- Amazon CloudWatch metrics for Amazon S3
- AWS CloudTrail support

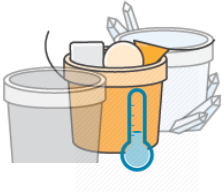


Amazon S3 bucket
limit increase

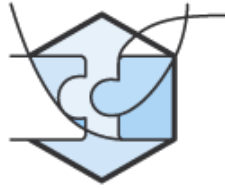


Read-after-write
consistency in all regions

Innovation for Amazon S3, continued...



Amazon S3
Standard-IA



Lifecycle policy



Transfer
Acceleration



Expired object delete
marker



Incomplete multipart
upload expiration

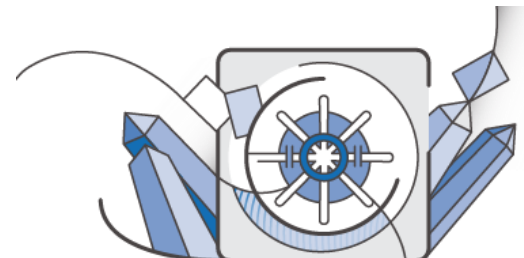
Choice of storage classes on Amazon S3



Standard



Standard - Infrequent Access



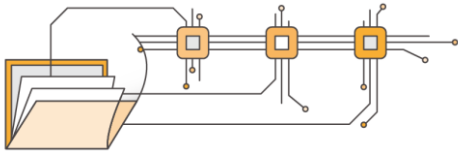
Amazon Glacier

Active data

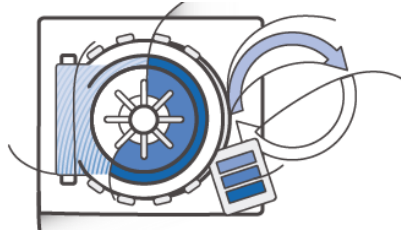
Infrequently accessed data

Archive data

Some use cases have different requirements



File sync and share
+
consumer file
storage



Backup and archive +
disaster recovery



Long retained
data

Standard-Infrequent Access storage



Durable

11 9s of durability



Available

Designed for
99.9% availability



High performance

Same throughput as
Amazon S3 Standard storage



Easy to use

- No impact on user experience
- Simple REST API
- Single bucket



Secure

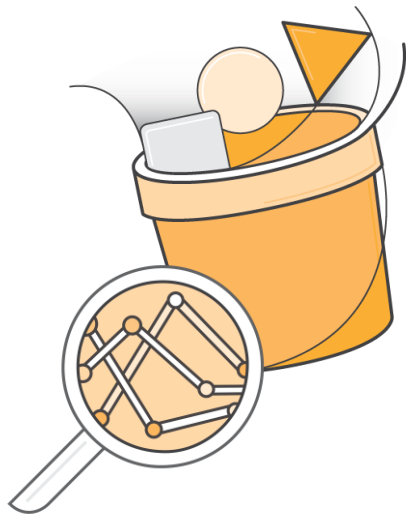
- Server-side encryption
- Use your encryption keys
- KMS-managed encryption keys



Integrated

- Lifecycle management
- Versioning
- Event notifications
- Metrics

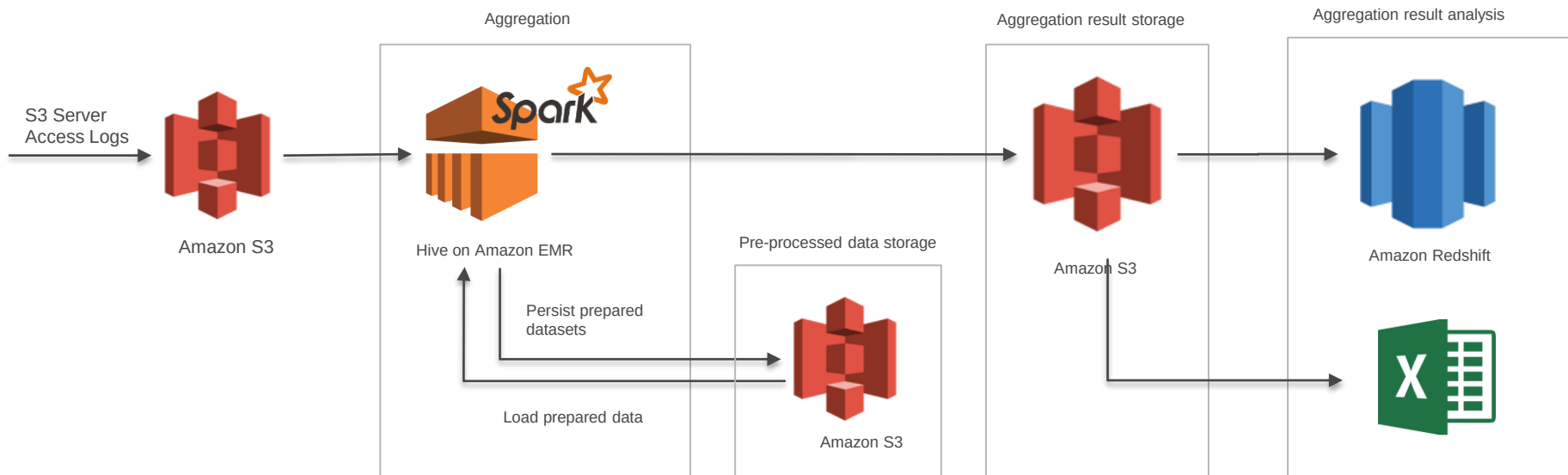
Understand your cloud storage

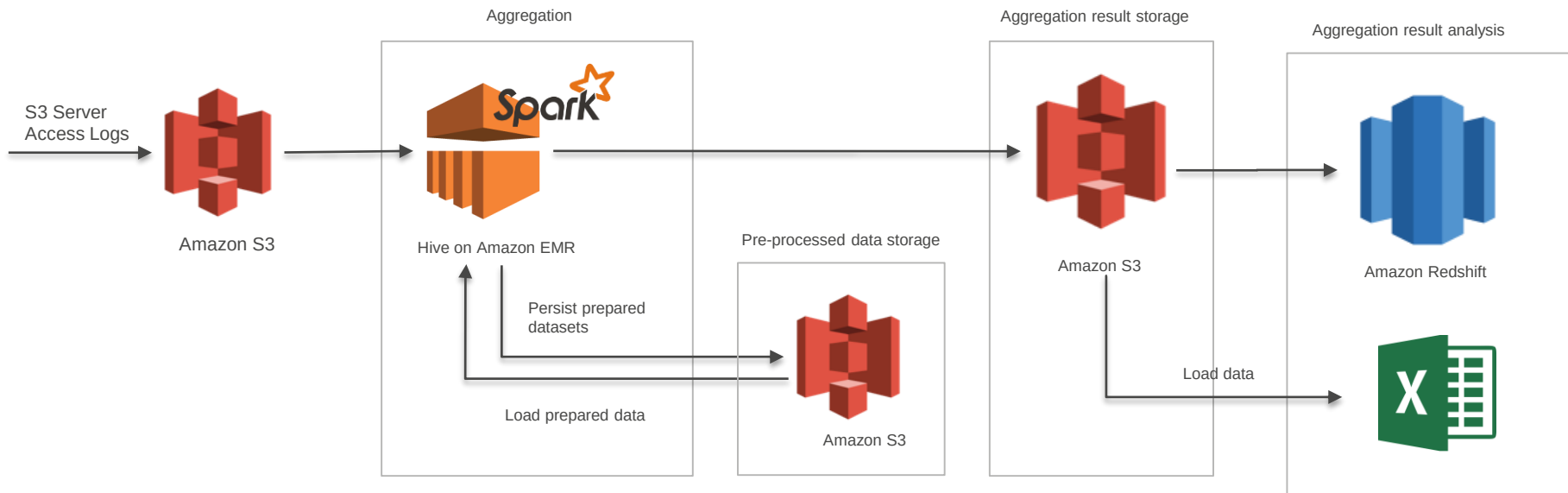


- Help me understand usage patterns
- Help me reduce cost
 - Which of my prefixes has infrequently accessed data?
 - How is performance changing for my bucket?

Understand your cloud storage

- Aggregate S3 server access logs
- Leverage Amazon EMR with Spark to aggregate at scale





1. Enable Access
Logs

2. Create EMR Cluster
3. Spark code to
aggregate logs
4. Submit code to EMR

5. Persist
interim
results on S3

6. Persist
final results
on S3

7. Visualize Data



DEMO

Understanding your cloud storage



S3 Management Console

https://console.aws.amazon.com/s3/home?region=us-west-2

mac rename file

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AWS

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S3

EMR

Lambda

Redshift

IAM

Edit

ConsoleAdmin @ s3-chicago-su...GlobalSupport

Create BucketActions

NonePropertiesTransfers

All Buckets (3)

Name
aws-logs-352228710819-us-east-1
s3-summit-emr
s3-summit-example

Bucket: s3-summit-example

Region: Oregon

Creation Date: Fri Apr 15 08:14:53 GMT-700 2016

Owner: s3-summit-demos

Permissions

Static Website Hosting

Logging

You can enable logging to track requests for access to your bucket. [Learn more.](#)

Enabled: ☒

Target Bucket: s3-summit-emr

Target Prefix: s3-access-logs

SaveCancel

Events

Versioning

Lifecycle

Cross-Region Replication

Tags

Requester Pays

Feedback

English

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S3 Management Console

https://console.aws.amazon.com/s3/home?region=us-west-2&bucket=s3-summit-emr&prefix=

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UploadCreate FolderActionsVersions:HideShow

Search by prefixNonePropertiesTransfers

All Buckets / s3-summit-emr

	Name	Storage Class	Size	Last Modified
☐	jars	--	--	--
☐	logs	--	--	--
☐	output	--	--	--
☐	s3-access-logs	--	--	--
☐	scripts	--	--	--

Feedback

English

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1b285544915b960a2ae2b1fd27431657c2d72553e78d995d76975dffc020cc43 s3-summit-example [27/Mar/2016:00:33:59 +0000] 10.1.1.1 arn:aws:iam::123412341234:user/summit 8441739623DDDB39 REST.GET.OBJECT et/gux/adipiscing/a.txt "PUT test1/test2/test3/object.txt HTTP/1.1" 200 - - 123 123 123 "-" "Mozilla/5.0 (Android 4.4; Mobile; rv:41.0) Gecko/41.0 Firefox/41.0" -

1b285544915b960a2ae2b1fd27431657c2d72553e78d995d76975dffc020cc43 s3-summit-example [27/Mar/2016:01:18:10 +0000] 10.1.1.1 arn:aws:iam::123412341234:user/summit 8441739623DDDB39 REST.PUT.OBJECT amet/sit/et/b.doc "PUT test1/test2/test3/object.txt HTTP/1.1" 200 - - 123 123 123 "-" "Mozilla/5.0 (Android 4.4; Mobile; rv:41.0) Gecko/41.0 Firefox/41.0" -

1b285544915b960a2ae2b1fd27431657c2d72553e78d995d76975dffc020cc43 s3-summit-example [27/Mar/2016:01:23:18 +0000] 10.1.1.1 arn:aws:iam::123412341234:user/summit 8441739623DDDB39 REST.DELETE.OBJECT consectetur/amet/gux/b.doc "PUT test1/test2/test3/object.txt HTTP/1.1" 204 - - 123 123 123 "-" "Mozilla/5.0 (Android 4.4; Mobile; rv:41.0) Gecko/41.0 Firefox/41.0" -

1b285544915b960a2ae2b1fd27431657c2d72553e78d995d76975dffc020cc43 s3-summit-example [27/Mar/2016:01:05:54 +0000] 10.1.1.1 arn:aws:iam::123412341234:user/summit 8441739623DDDB39 REST.PUT.OBJECT baz/unim/bar/a.txt "PUT test1/test2/test3/object.txt HTTP/1.1" 200 - - 123 123 123 "-" "Mozilla/5.0 (Android 4.4; Mobile; rv:41.0) Gecko/41.0 Firefox/41.0" -

1b285544915b960a2ae2b1fd27431657c2d72553e78d995d76975dffc020cc43 s3-summit-example [27/Mar/2016:01:17:46 +0000] 10.1.1.1 arn:aws:iam::123412341234:user/summit 8441739623DDDB39 REST.GET.OBJECT sit/bar/a.txt "PUT test1/test2/test3/object.txt HTTP/1.1" 200 - - 123 123 123 "-" "Mozilla/5.0 (Android 4.4; Mobile; rv:41.0) Gecko/41.0 Firefox/41.0" -

1b285544915b960a2ae2b1fd27431657c2d72553e78d995d76975dffc020cc43 s3-summit-example [27/Mar/2016:00:40:10 +0000] 10.1.1.1 arn:aws:iam::123412341234:user/summit 8441739623DDDB39 REST.DELETE.OBJECT adipiscing/baz/amet/b.doc "PUT test1/test2/test3/object.txt HTTP/1.1" 204 - - 123 123 123 "-" "Mozilla/5.0 (Android 4.4; Mobile; rv:41.0) Gecko/41.0 Firefox/41.0" -

1b285544915b960a2ae2b1fd27431657c2d72553e78d995d76975dffc020cc43 s3-summit-example [27/Mar/2016:00:28:30 +0000] 10.1.1.1 arn:aws:iam::123412341234:user/summit 8441739623DDDB39 REST.GET.OBJECT adipiscing/foo/elit/a.txt "PUT test1/test2/test3/object.txt HTTP/1.1" 200 - - 123 123 123 "-" "Mozilla/5.0 (Android 4.4; Mobile; rv:41.0) Gecko/41.0 Firefox/41.0" -

AWS Elastic MapReduce M...

https://us-west-2.console.aws.amazon.com/elasticmapreduce/home?region=us-west-2#quick-create:

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Oregon

Support

Elastic MapReduce

Create Cluster

EMR Help

Create Cluster - Quick Options

[Go to advanced options](#)

General Configuration

Cluster name

Example Cluster

☒ Logging

S3 folder

s3://aws-logs-352228710819-us-west-2/elasticmapredu

Launch mode

☒ Cluster

☐ Step execution

Software configuration

Vendor

☒ Amazon

☐ MapR

Release

emr-4.5.0

Applications

☒ All Applications: Ganglia 3.7.2, Hadoop 2.7.2, Hive 1.0.0, Hue 3.7.1, Mahout 0.11.1, Pig 0.14.0, and Spark 1.6.1

☐ Core Hadoop: Hadoop 2.7.2 with Ganglia 3.7.2, Hive 1.0.0, and Pig 0.14.0

☐ Presto-Sandbox: Presto 0.140 with Hadoop 2.7.2 HDFS and Hive 1.0.0 Metastore

☐ Spark: Spark 1.6.1 on Hadoop 2.7.2 YARN with Ganglia 3.7.2

AWS Elastic MapReduce M... x

https://us-west-2.console.aws.amazon.com/elasticmapreduce/home?region=us-west-2#quick-create: Search

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Applications: Ganglia 3.7.2, Hadoop 2.7.2, Hive 1.0.0, Hue 3.7.1, Mahout 0.11.1, Pig 0.14.0, and Spark 1.6.1

☐ Core Hadoop: Hadoop 2.7.2 with Ganglia 3.7.2, Hive 1.0.0, and Pig 0.14.0

☐ Presto-Sandbox: Presto 0.140 with Hadoop 2.7.2 HDFS and Hive 1.0.0 Metastore

☐ Spark: Spark 1.6.1 on Hadoop 2.7.2 YARN with Ganglia 3.7.2

Hardware configuration

Instance type

r3.xlarge

Number of instances

5

 (1 master and 4 core nodes)

Security and access

EC2 key pair

S3SummitDemo_us-west-2

Learn how to create an EC2 key pair.

Permissions ☒ Default ☐ Custom
Use default IAM roles. If roles are not present, they will be automatically created for you with managed policies for automatic policy updates.

EMR role [EMR_DefaultRole](#) ⓘ

EC2 instance profile [EMR_EC2_DefaultRole](#) ⓘ

Cancel

Create cluster

Feedback

English

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⌵

AWS Elastic MapReduce M...

✕

⌵

⏪

https://us-west-2.console.aws.amazon.com/elasticmapreduce/home?region=us-west-2#cluster-details:j-2K726XK

🔍 Chicago metra train schedule

☆

📁

⬇️

🏠

📶

😊

☰

Amazon.com

AWS Console

Summit Console ...

📦 AWS

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Services

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📁 S3

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📦 EMR

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📦 Lambda

⌵

📦 Redshift

⌵

📦 IAM

⌵

Edit

⌵

ConsoleAdmin @ s3-chicago-su...

⌵

Oregon

⌵

Support

⌵

Elastic MapReduce

⌵

Cluster List

>

Cluster Details

EMR Help

Add step

Resize

Clone

Cluster: S3 Chicago Summit

Connections: [Enable Web](#)

Master public DNS: ec2-52-32-

Tags: -- [View All](#)

Summary

ID: j-2K726XKJG03O3

Creation date: 2016-04-15 07:51 (UTC-7)

Elapsed time: 2 days, 11 hours

Auto-terminate: No

Termination protection: Off [Change](#)

Monitoring

Hardware

Steps

Add Step

✕

Step type

Spark application

⌵

Name

Prefix Aggregation

Deploy mode

Cluster

⌵

Run your driver on a slave node (cluster mode) or on the master node as an external client (client mode).

Spark-submit options

--class s3.summit.Main

Specify other options for spark-submit.

Application location*

s3://s3-summit-emr/jars/S32016SummitDemo-1.0.jar

📁 Path to a JAR with your application and dependencies (client deploy mode only supports a local path).

Arguments

--input s3://s3-summit-emr/s3-access-logs/ --output s3://s3-summit-emr/output/prefixAggregates

Specify optional arguments for your application.

Action on failure

Continue

⌵

What to do if the step fails.

Cancel

Add

Add step Resize Clone Terminate AWS CLI export

Cluster: S3 Chicago Summit Waiting Cluster ready after last step completed.

Connections: [Enable Web Connection](#) – Hue, Spark History Server, Ganglia, Resource Manager ... (View All)

Master public DNS: [ec2-52-32-113-237.us-west-2.compute.amazonaws.com](#) [SSH](#)

Tags: -- [View All / Edit](#)

Summary

ID: j-2K726XKJG03O3

Creation date: 2016-04-15 07:51 (UTC-7)

Elapsed time: 3 days, 9 hours

Auto-terminate: No

Termination protection: Off [Change](#)

Configuration Details

Release label: emr-4.5.0

Hadoop distribution: Amazon 2.7.2

Applications: Ganglia 3.7.2, Hive 1.0.0, Hue 3.7.1, Mahout 0.11.1, Pig 0.14.0, Spark 1.6.1

Log URI: --

EMRFS consistent view: Disabled

Network and Hardware

Availability zone: us-west-2c

Subnet ID: [subnet-97f6f4ce](#)

Master: Running 1 r3.xlarge

Core: Running 4 r3.xlarge

Task: --

Security and Access

Key name: S3SummitDemo_us-west-2

EC2 instance profile: EMR_EC2_DefaultRole

EMR role: EMR_DefaultRole

Visible to all users: All [Change](#)

Security groups for (ElasticMapReduce-Master): [sg-35154a52](#)

Security groups for (ElasticMapReduce-slave): [sg-33154a54](#)

Core & Task:

► Monitoring

► Hardware

Termination protection: Off [Change](#)

Log URI: --
EMRFS Disabled
consistent view:

Core: **Running** 4 r3.xlarge
Task: --

EMR role: EMR_DefaultRole
Visible to all users: All [Change](#)
Security groups for (ElasticMapReduce-Master: master) sg-35154a52
Security groups for (ElasticMapReduce-slave) sg-33154a54
Core & Task:

► Monitoring

► Hardware

▼ Steps

[Add step](#) [Clone step](#)

Steps

[View all interactive jobs](#) | [View all jobs](#)

Filter: All steps Filter steps ...		1 step (all loaded)					
	ID	Name	Status	Start time (UTC-7)	Elapsed time	Log files	Action
☐	s-Y0SWNL6ESESV	Setup hadoop debugging	Completed	2016-04-15 08:02 (UTC-7)	2 seconds	Logging not configured	View

► Configurations

► Bootstrap Actions

```

28 SparkConf sparkConf = new SparkConf().setAppName("S3EmrDemo");
29 try(JavaSparkContext jsc = new JavaSparkContext(sparkConf)){
30
31     /*
32      * Pull the raw data down from S3.
33      * Let Spark choose a partition size, and thus the number
34      * of partitions, for us.
35      */
36     JavaRDD<String> rawData = jsc
37         .textFile(cmd.getOptionValue("input"), 200);
38
39     /*
40      * Parse the data and filter records that failed parsing
41      * This would be a good opportunity to log information about these
42      * parsing failures to improve the parsing logic for future runs.
43      */
44     JavaRDD<S3AccessLogRecord> nonNullParsedData =
45         rawData
46             .map(new S3AccessLogParser())
47             .filter(Objects::nonNull)
48             .cache();
49
50     /*
51      * Persist the parsed and filtered data to S3.
52      * In this was we make the data easily accessible to other
53      * applications, possibly not running on the same cluster. In
54      * production we'd choose to use a more general serialization
55      * format.
56      */
57     nonNullParsedData.saveAsObjectFile(generateS3LocationForIntermediateResults());
58
59     /*
60      * Apply aggregation logic to parsed data.
61      * This process will aggregate our parsed data and upload it to the
62      * location specified by the 'output' parameter to the application.
63      */
64     new PrefixAggregation(cmd.getOptionValue("output")).apply(nonNullParsedData);
65

```

Main Spark app

Persist pre-processed data
in S3

```

35 public void apply(JavaRDD<S3AccessLogRecord> dataset) {
36     /*
37      * Process individual log entries, extracting the prefix and other interesting information,
38      * i.e. Prefix | Timestamp (by hour) | S3 Operation | Status code.
39      */
40     JavaRDD<PrefixRecord> aggByPrefix =
41         dataset.map(new MapS3AccessRecordToPrefixRecord());
42
43     /*
44      * For this report we're going to aggregate the records using the above format
45      * as a primary key. There's lots of other ways to accomplish this,
46      * e.g. Spark's DataFrames.
47      */
48     aggByPrefix
49
50         /*
51          * Begin by associating (mapping) each record to the value 1L
52          */
53         .mapToPair(record -> new Tuple2<>(record, new AtomicLong(1)))
54
55         /*
56          * Reduce the data, accumulating the count for each equivalent record
57          */
58         .reduceByKey((left, right) -> {
59             left.getAndAdd(right.get());
60             return left;})
61
62         /*
63          * Serialize as csv for ease of use upstream
64          */
65         .map(recordTuple -> recordTuple._1.toString() + "," + recordTuple._2)
66
67         /*
68          * Persist the aggregated results to S3
69          */
70         .saveAsTextFile(outputLocation);
71
72 }

```

Prefix aggregation

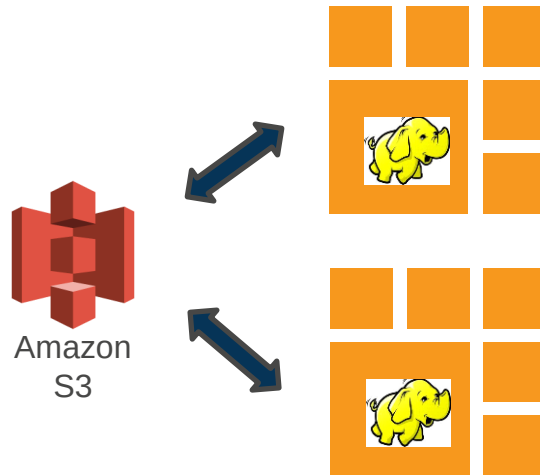
Persist result in S3

Amazon S3 as your persistent data store

Separate compute and storage

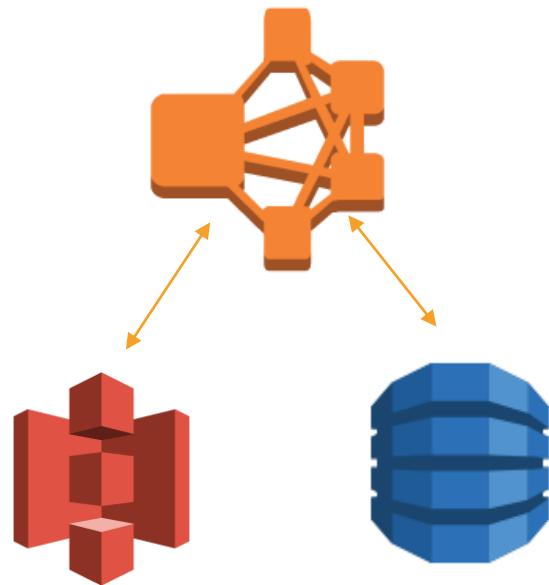
Resize and shut down Amazon EMR clusters with no data loss

Point multiple Amazon EMR clusters at the same data in Amazon S3



EMRFS makes it easier to use Amazon S3

- Read-after-write consistency
- Very fast list operations
- Error handling options
- Support for Amazon S3 encryption
- Transparent to applications: `s3://`



S3 Management Console

https://console.aws.amazon.com/s3/home?region=us-west-2#&bucket=s3-summit-emr&prefix=intermediate

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Amazon.com

AWS Console

Summit Console ...

AWS

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Redshift

IAM

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Global

Support

Upload

Create Folder

Actions

Versions: Hide Show

Search by prefix

None

Properties

Transfers

All Buckets

/ s3-summit-emr

/ intermediate

/ prefixAggregation

	Name	Storage Class	Size	Last Modified
☐	_SUCCESS	Standard	0 bytes	Fri Apr 15 22:16:25 GMT-700 2016
☐	part-00000	Standard	5.2 MB	Fri Apr 15 22:14:00 GMT-700 2016
☐	part-00001	Standard	5.8 MB	Fri Apr 15 22:14:00 GMT-700 2016
☐	part-00002	Standard	6.4 MB	Fri Apr 15 22:14:00 GMT-700 2016
☐	part-00003	Standard	7 MB	Fri Apr 15 22:14:00 GMT-700 2016
☐	part-00004	Standard	7.6 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00005	Standard	8.1 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00006	Standard	8.6 MB	Fri Apr 15 22:14:07 GMT-700 2016
☐	part-00007	Standard	9.1 MB	Fri Apr 15 22:14:07 GMT-700 2016
☐	part-00008	Standard	9.5 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00009	Standard	9.8 MB	Fri Apr 15 22:14:07 GMT-700 2016
☐	part-00010	Standard	10 MB	Fri Apr 15 22:14:07 GMT-700 2016
☐	part-00011	Standard	10.2 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00012	Standard	10.3 MB	Fri Apr 15 22:14:08 GMT-700 2016
☐	part-00013	Standard	10.4 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00014	Standard	10.3 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00015	Standard	10.2 MB	Fri Apr 15 22:14:07 GMT-700 2016
☐	part-00016	Standard	10 MB	Fri Apr 15 22:14:07 GMT-700 2016
☐	part-00017	Standard	9.7 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00018	Standard	9.4 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00019	Standard	9 MB	Fri Apr 15 22:14:06 GMT-700 2016
☐	part-00020	Standard	8.5 MB	Fri Apr 15 22:14:06 GMT-700 2016

Feedback

English

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	A	B	C	D	E
	Prefix	Timestamp	Operation	Status	Count
1					
2	/amet/bar/	4/14/16 21:00	REST.GET.OBJECT	500	5
3	/elit/sed/	4/12/16 3:00	REST.DELETE.OBJECT	503	15
4	/foo/	4/7/16 18:00	REST.PUT.OBJECT	403	130
5	/unim/adipis	3/27/16 1:00	REST.DELETE.OBJECT	500	14
6	/sit/consecte	4/4/16 14:00	REST.GET.OBJECT	400	3
7	/foo/	4/8/16 23:00	REST.PUT.OBJECT	403	69
8	/do/bar/	4/3/16 5:00	REST.GET.OBJECT	403	1
9	/adipiscing/i	3/31/16 17:00	REST.PUT.OBJECT	403	2
10	/bar/foo/	4/19/16 16:00	REST.DELETE.OBJECT	400	212
11	/unim/adipis	3/27/16 5:00	REST.DELETE.OBJECT	500	30
12	/unim/do/	3/27/16 20:00	REST.GET.OBJECT	403	7
13	/adipiscing/i	4/4/16 9:00	REST.PUT.OBJECT	403	1
14	/elit/baz/	4/2/16 10:00	REST.GET.OBJECT	500	2
15	/bar/foo/	4/17/16 21:00	REST.DELETE.OBJECT	400	110
16	/bar/foo/	4/18/16 0:00	REST.DELETE.OBJECT	400	100
17	/amet/bar/	4/22/16 20:00	REST.GET.OBJECT	500	5
18	/elit/foo/	4/21/16 16:00	REST.DELETE.OBJECT	403	1
19	/amet/bar/	4/17/16 9:00	REST.GET.OBJECT	500	4
20	/qux/baz/	4/1/16 22:00	REST.GET.OBJECT	400	2
21	/unim/adipis	3/27/16 17:00	REST.DELETE.OBJECT	500	16
22	/et/adipiscin	4/22/16 3:00	REST.DELETE.OBJECT	503	2
23	/qux/baz/	3/29/16 11:00	REST.GET.OBJECT	400	14
24	/et/baz/	3/27/16 6:00	REST.DELETE.OBJECT	400	6
25	/foo/	3/28/16 19:00	REST.PUT.OBJECT	403	24
26	/elit/sed/	4/18/16 3:00	REST.DELETE.OBJECT	503	4
27	/elit/baz/	4/6/16 15:00	REST.GET.OBJECT	500	1
28	/adipiscing/i	4/9/16 12:00	REST.GET.OBJECT	503	12
29	/sit/consecte	4/4/16 19:00	REST.GET.OBJECT	400	3
30	/et/baz/	3/28/16 6:00	REST.DELETE.OBJECT	400	11
31	/adipiscing/i	4/9/16 5:00	REST.GET.OBJECT	503	8
32	/amet/bar/	4/25/16 5:00	REST.GET.OBJECT	500	4
33	/elit/sed/	4/11/16 19:00	REST.DELETE.OBJECT	503	19
34	/adipiscing/i	4/2/16 20:00	REST.PUT.OBJECT	403	1
35	/sed/sit/	3/28/16 18:00	REST.DELETE.OBJECT	200	22
36	/bar/foo/	4/11/16 16:00	REST.DELETE.OBJECT	400	103

Excel status bar: Pivot1, AggregatedData.csv, Chart 1, Normal View, Ready.

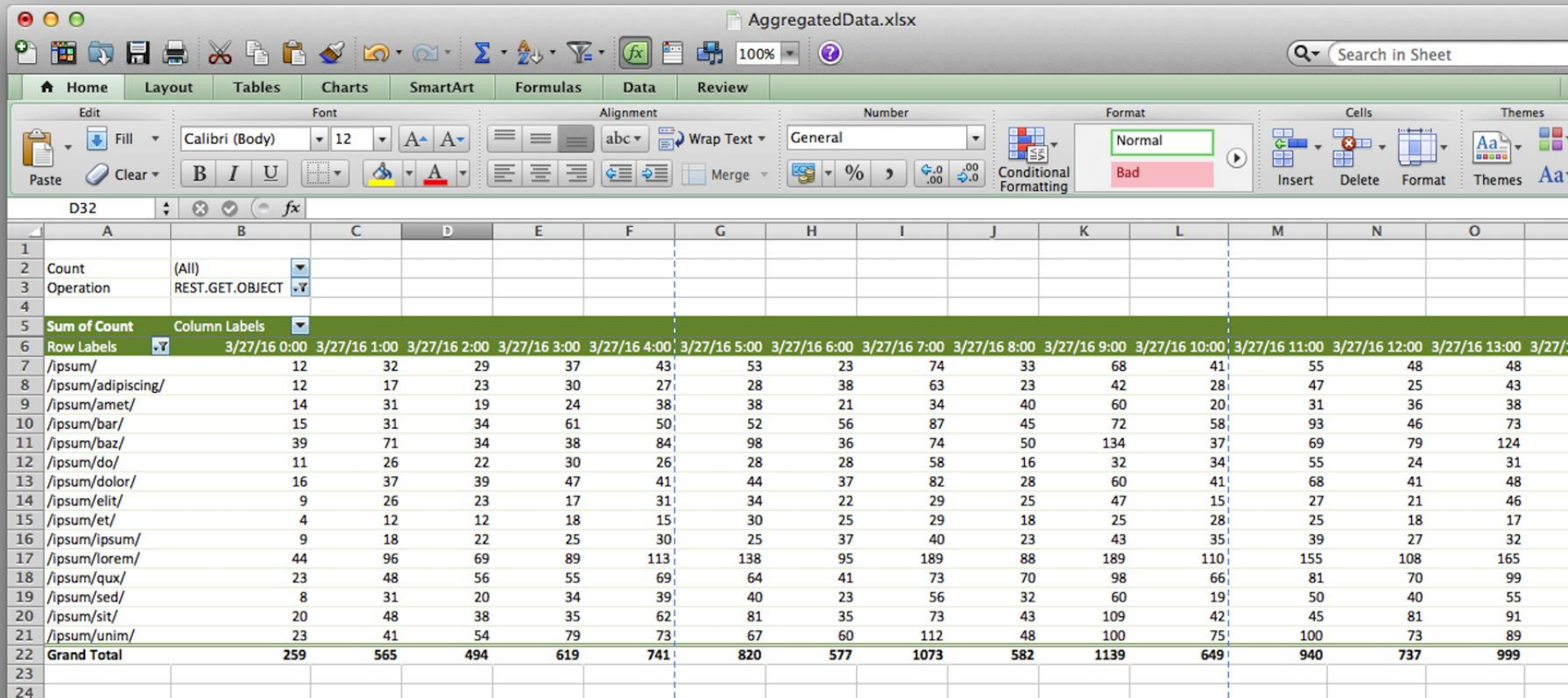
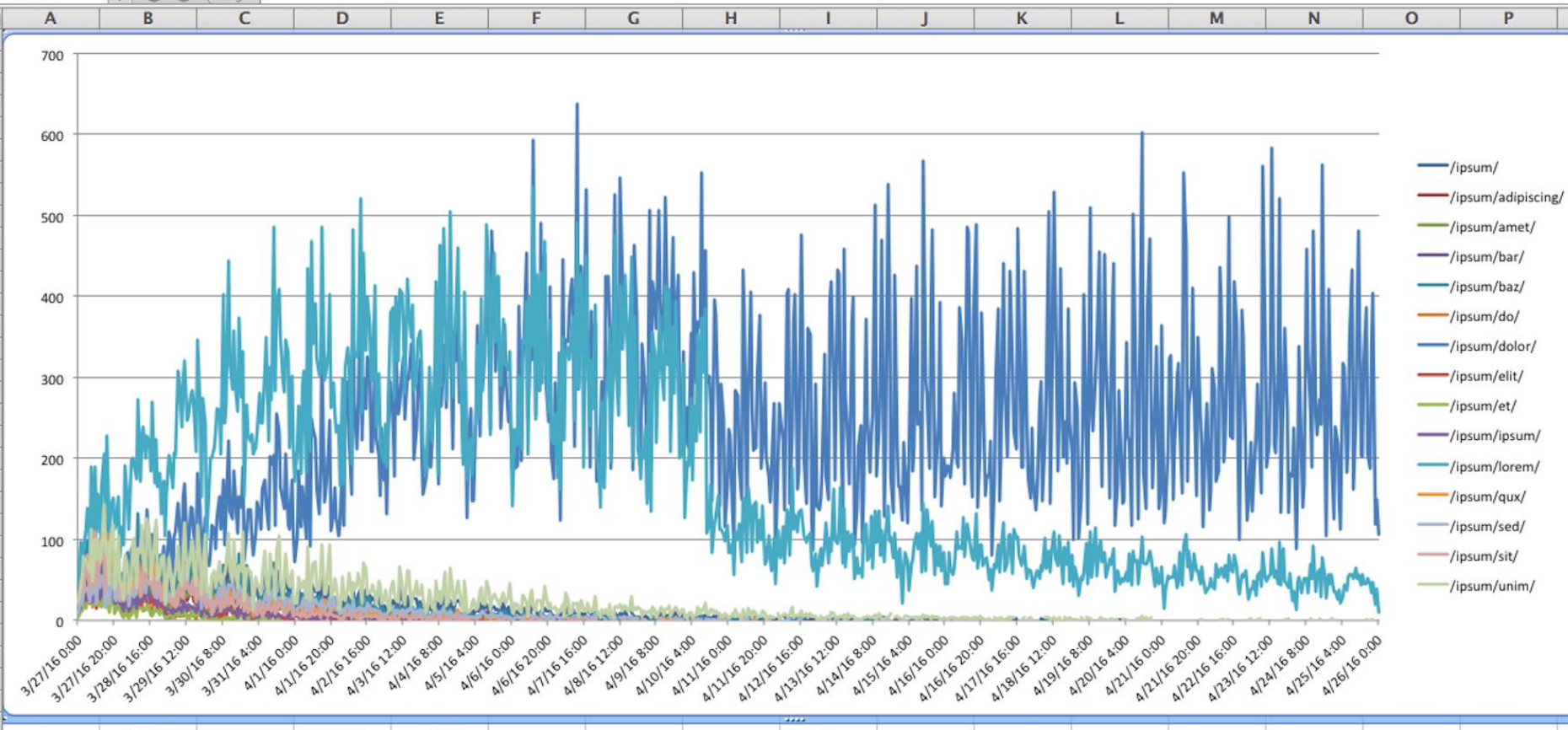
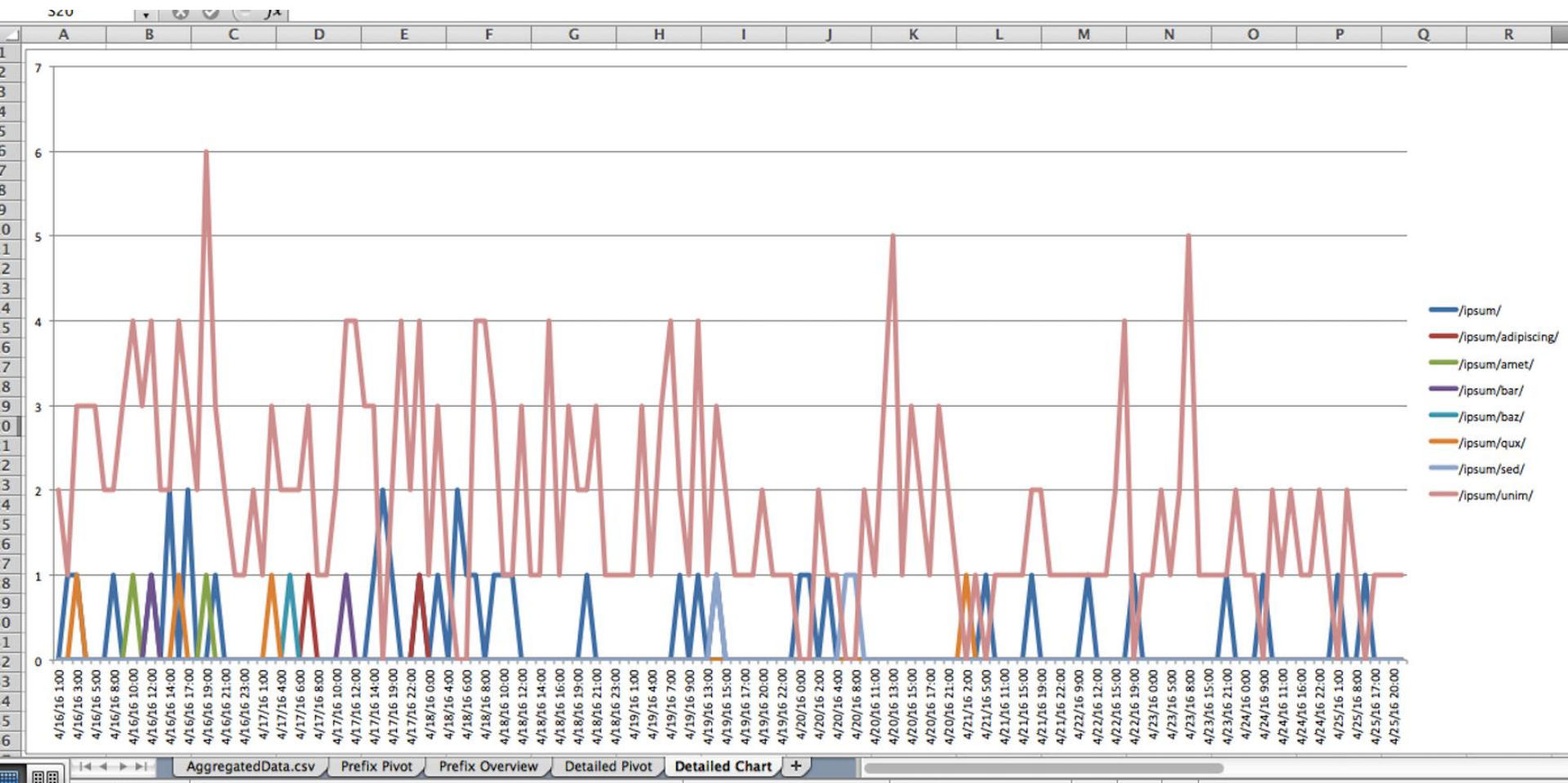


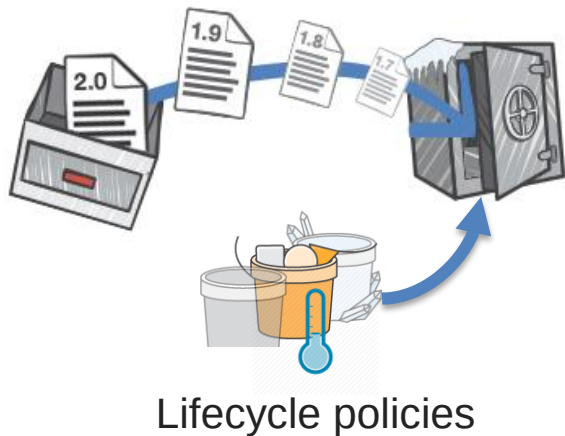
Chart 1





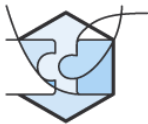
Management policies

Lifecycle policies



- Automatic tiering and cost controls
- Includes two possible actions:
 - Transition: archives to Standard-IA or Amazon Glacier after specified time
 - Expiration: deletes objects after specified time
- Allows for actions to be combined
- Set policies at the prefix level

Standard-Infrequent Access storage



Integrated: Lifecycle management



Standard - Infrequent Access

- Transition Standard to Standard-IA
- Transition Standard-IA to Amazon Glacier storage
- Expiration lifecycle policy
- Versioning support
- Directly PUT to Standard-IA

Standard-Infrequent Access storage

Lifecycle policy

```
<LifecycleConfiguration>
  <Rule>
    <ID>sample-rule</ID>
    <Prefix>documents/</Prefix>
    <Status>Enabled</Status>
    <Transition>
      <Days>30</Days>
      <StorageClass>STANDARD-IA</StorageClass>
    </Transition>
    <Transition>
      <Days>365</Days>
      <StorageClass>GLACIER</StorageClass>
    </Transition>
  </Rule>
</LifecycleConfiguration>
```

Standard Storage -> Standard-IA

Standard-Infrequent Access storage

Lifecycle Policy

```
<LifecycleConfiguration>
  <Rule>
    <ID>sample-rule</ID>
    <Prefix>documents/</Prefix>
    <Status>Enabled</Status>
    <Transition>
      <Days>30</Days>
      <StorageClass>STANDARD-IA</StorageClass>
    </Transition>
    <Transition>
      <Days>365</Days>
      <StorageClass>GLACIER</StorageClass>
    </Transition>
  </Rule>
</LifecycleConfiguration>
```

Standard Storage -> Standard-IA

Standard-IA Storage -> Amazon Glacier

Versioning S3 buckets



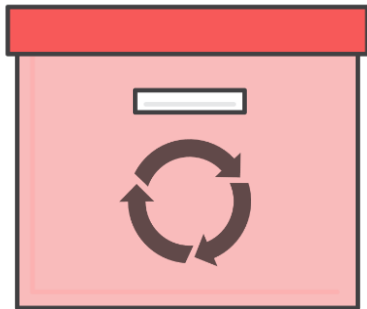
Versioning

- Protects from accidental overwrites and deletes
- New version with every upload
- Easy retrieval of deleted objects and roll back
- Three states of an Amazon S3 bucket
 - Default – Unversioned
 - Versioning-enabled
 - Versioning-suspended



Best Practice

Versioning + lifecycle policies



Versioning



Recycle bin



Lifecycle
policies



Automatic
cleaning

Expired object delete marker policy



Expired object delete
marker

- Deleting a versioned object makes a delete marker the current version of the object
- No storage charge for delete marker
- Removing delete marker can improve list performance
- Lifecycle policy to automatically remove the current version delete marker when previous versions of the object no longer exist

Expired object delete marker policy

Example lifecycle policy to remove current versions

```
<LifecycleConfiguration>
  <Rule>
    ...
    <Expiration>
      <Days>60</Days>
    </Expiration>
    <NoncurrentVersionExpiration>
      <NoncurrentDays>30</NoncurrentDays>
    </NoncurrentVersionExpiration>
  </Rule>
</LifecycleConfiguration>
```

Leverage lifecycle to expire current and non-current versions

S3 Lifecycle will automatically remove any expired object delete markers

Expired object delete marker policy

Example lifecycle policy for non-current version expiration

```
<LifecycleConfiguration>
  <Rule>
    ...
    <Expiration>
      <ExpiredObjectDeleteMarker>true</ExpiredObjectDeleteMarker>
    </Expiration>
    <NoncurrentVersionExpiration>
      <NoncurrentDays>30</NoncurrentDays>
    </NoncurrentVersionExpiration>
  </Rule>
</LifecycleConfiguration>
```

ExpiredObjectDeleteMarker element removes expired object delete markers.

Lifecycle configuration with **NoncurrentVersionExpiration** action removes all the noncurrent versions,



DEMO

Expired object delete marker policy





Tip: Restricting deletes

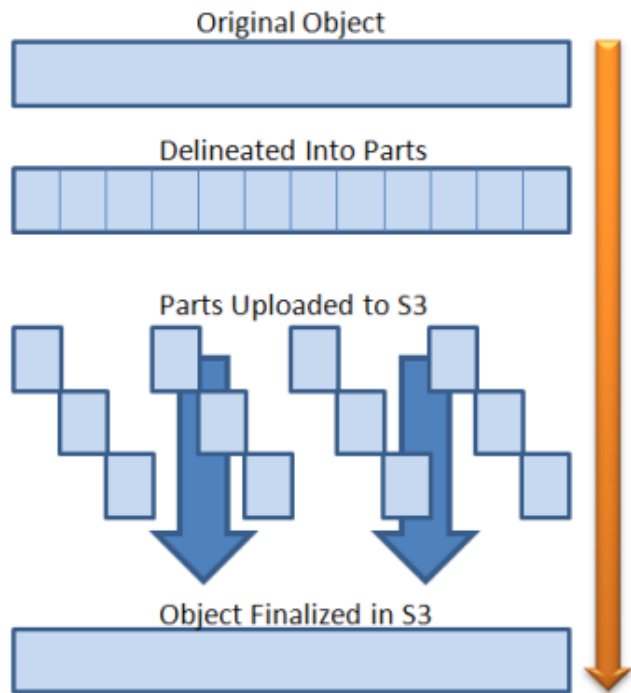
- Bucket policies can restrict deletes
- For additional security, enable MFA (multi-factor authentication) delete, which requires additional authentication to:
 - Change the versioning state of your bucket
 - Permanently delete an object version
- MFA delete requires both your security credentials and a code from an approved authentication device



Best Practice

Performance optimization for S3

Parallelizing PUTs with multipart uploads



- Increase aggregate throughput by parallelizing PUTs on high-bandwidth networks
 - Move the bottleneck to the network where it belongs
- Increase resiliency to network errors; fewer large restarts on error-prone networks



Best Practice

Multipart upload provides parallelism

- Allows *faster*, more *flexible* uploads
- Allows you to upload a single object as a *set of parts*
- Upon upload, Amazon S3 then *presents all parts as a single object*
- Enables *parallel* uploads, *pausing* and *resuming* an object upload and *starting uploads before you know the total object size*

Incomplete multipart upload expiration policy



Incomplete multipart
upload expiration

- Multipart upload feature improves PUT performance
- Partial upload does not appear in bucket list
- Partial upload does incur storage charges
- Set a lifecycle policy to automatically expire incomplete multipart uploads after a predefined number of days

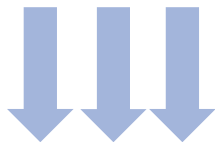
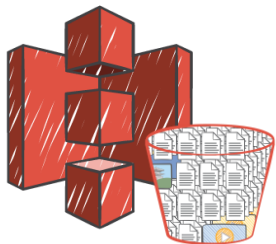
Incomplete multipart upload expiration policy

Example lifecycle policy

```
<LifecycleConfiguration>
  <Rule>
    <ID>sample-rule</ID>
    <Prefix>SomeKeyPrefix</Prefix>
    <Status>rule-status</Status>
    <AbortIncompleteMultipartUpload>
      <DaysAfterInitiation>7</DaysAfterInitiation>
    </AbortIncompleteMultipartUpload>
  </Rule>
</LifecycleConfiguration>
```

Abort incomplete multipart uploads
seven days after initiation

Parallelize your GETs



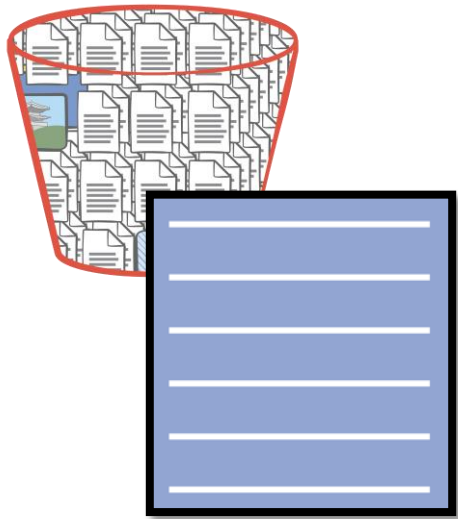
Amazon EC2

- Use range-based GETs to get multithreaded performance when downloading objects
- Compensates for unreliable networks
- Benefits of multithreaded parallelism



Best Practice

Parallelizing LIST



Parallelize LIST when you need a sequential list of your keys

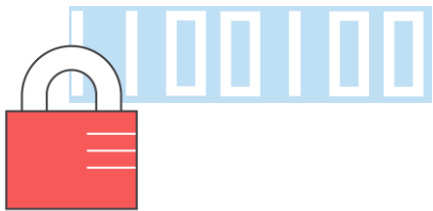
Secondary index to get a faster alternative to LIST

- Sorting by metadata
- Search ability
- Objects by timestamp



Best Practice

SSL best practices to optimize performance



- Use the SDKs!!
- EC2 instance types
 - AES-NI hardware acceleration (`cat /proc/cpuinfo`)
 - Threads can work against you (finite network capacity)
- Timeouts
 - Connection pooling
 - Perform keep-alives to avoid handshake



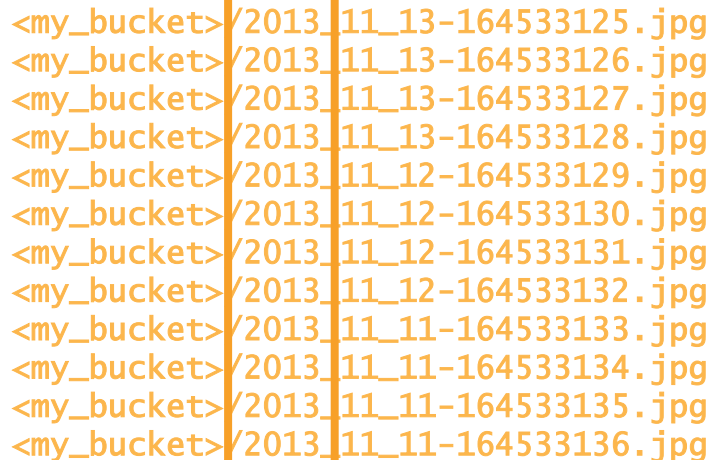
Best Practice

Distributing key names

Use a key-naming scheme with randomness at the beginning for high TPS

- Most important if you regularly exceed 100 TPS on a bucket
- Avoid starting with a date
- Consider adding a hash or reversed timestamp (ssmmhhddmmyy)

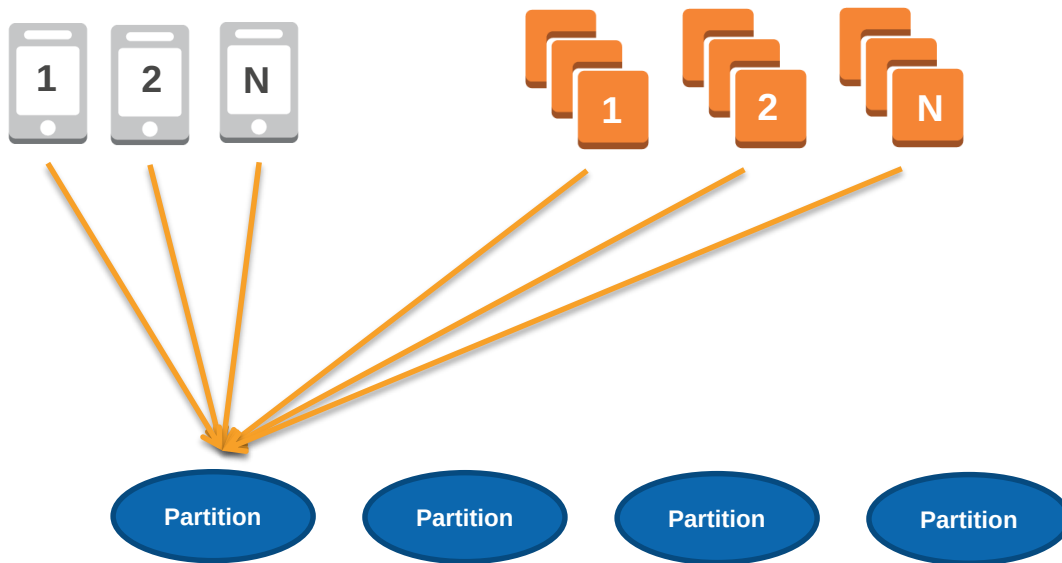
Don't do this...



```
<my_bucket>/2013_11_13-164533125.jpg  
<my_bucket>/2013_11_13-164533126.jpg  
<my_bucket>/2013_11_13-164533127.jpg  
<my_bucket>/2013_11_13-164533128.jpg  
<my_bucket>/2013_11_12-164533129.jpg  
<my_bucket>/2013_11_12-164533130.jpg  
<my_bucket>/2013_11_12-164533131.jpg  
<my_bucket>/2013_11_12-164533132.jpg  
<my_bucket>/2013_11_11-164533133.jpg  
<my_bucket>/2013_11_11-164533134.jpg  
<my_bucket>/2013_11_11-164533135.jpg  
<my_bucket>/2013_11_11-164533136.jpg
```

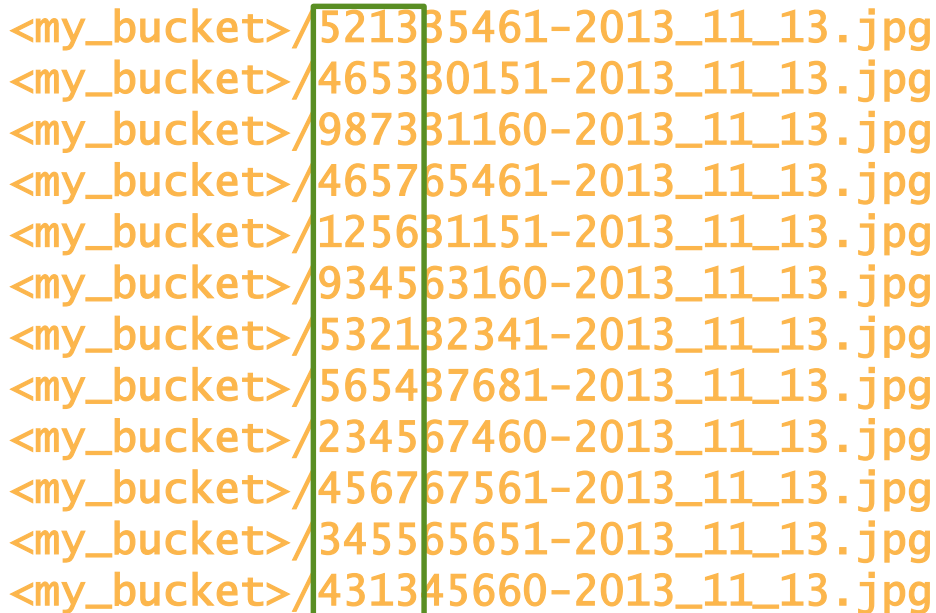
Distributing key names

...because this is going to happen



Distributing key names

Add randomness to the beginning of the key name...

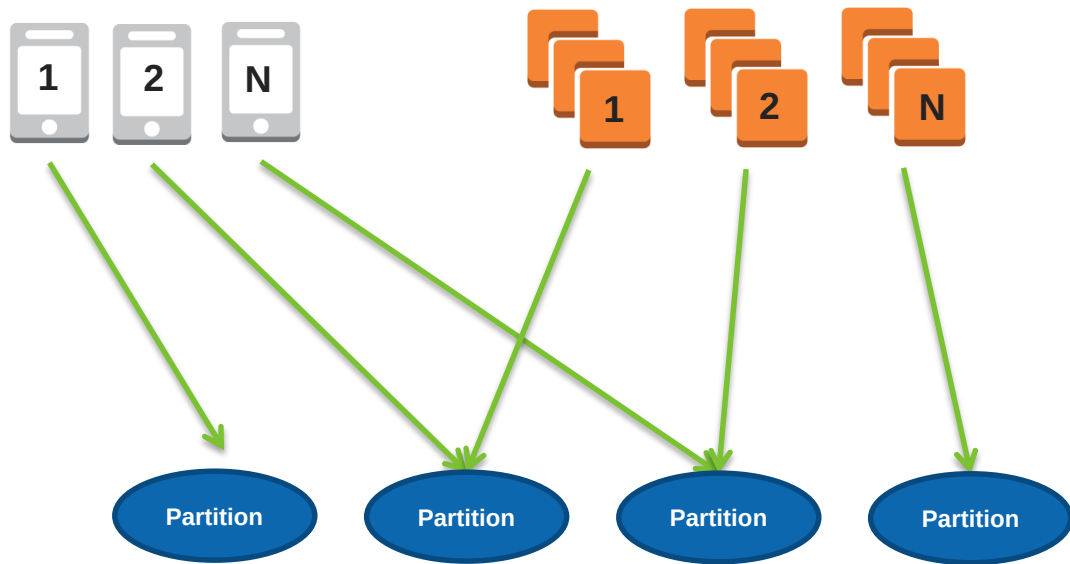


A list of 10 file paths, each starting with the prefix "<my_bucket>/". The paths are: <my_bucket>/521385461-2013_11_13.jpg, <my_bucket>/465380151-2013_11_13.jpg, <my_bucket>/987381160-2013_11_13.jpg, <my_bucket>/465765461-2013_11_13.jpg, <my_bucket>/125681151-2013_11_13.jpg, <my_bucket>/934563160-2013_11_13.jpg, <my_bucket>/532182341-2013_11_13.jpg, <my_bucket>/565487681-2013_11_13.jpg, <my_bucket>/234567460-2013_11_13.jpg, <my_bucket>/456767561-2013_11_13.jpg, <my_bucket>/345565651-2013_11_13.jpg, and <my_bucket>/431345660-2013_11_13.jpg. A green rectangular box highlights the random numeric prefix of each path, specifically the digits "521385461" through "431345660".

- <my_bucket>/521385461-2013_11_13.jpg
- <my_bucket>/465380151-2013_11_13.jpg
- <my_bucket>/987381160-2013_11_13.jpg
- <my_bucket>/465765461-2013_11_13.jpg
- <my_bucket>/125681151-2013_11_13.jpg
- <my_bucket>/934563160-2013_11_13.jpg
- <my_bucket>/532182341-2013_11_13.jpg
- <my_bucket>/565487681-2013_11_13.jpg
- <my_bucket>/234567460-2013_11_13.jpg
- <my_bucket>/456767561-2013_11_13.jpg
- <my_bucket>/345565651-2013_11_13.jpg
- <my_bucket>/431345660-2013_11_13.jpg

Distributing key names

...so your transactions can be distributed across the partitions



Best Practice

Other techniques for distributing key names



Store objects as a hash of their name

- add the original name as metadata
 - “deadmau5_mix.mp3” →
0aa316fb000eae52921aab1b4697424958a53ad9
- prepend key name with short hash
 - 0aa3-deadmau5_mix.mp3



(reverse)

- 5321354831-deadmau5_mix.mp3



Best Practice

Recap

- Amazon S3 Standard-Infrequent Access
- Using big data on Amazon S3 for analysis
- Amazon S3 management policies
- Versioning for Amazon S3
- Best practices and performance optimization for Amazon S3

